Rings and modular arithmetic solutions

1

A class a is a unit exactly when $\gcd(a, 8) = 1$. The units are $1, 3, 5, 7$. The remaining nonzero classes $2, 4, 6$ are zero divisors, as witnessed by

$$2 \cdot 4 \equiv 0, \qquad 4 \cdot 2 \equiv 0, \qquad 6 \cdot 4 \equiv 0 \pmod 8.$$

Since $3 \cdot 3 = 9 \equiv 1 \pmod 8$, the inverse of 3 is 3.

2

For $p, q \in \mathbb{Z}[x]$, substitution at a preserves addition and multiplication, and reduction modulo 12 also preserves both operations. The constant 1 maps to $[1]$, so this is a unital ring homomorphism. Constant polynomials attain every residue, hence the image is $\mathbb{Z}_{12}$.

Division by the monic polynomial $x - a$ over the integers gives a unique expression

$$p(x) = (x - a)q(x) + p(a).$$

Consequently $[p(a)]_{12} = 0$ exactly when $p(a) = 12k$, and

$$\ker ev_a = \{(x - a)q(x) + 12k : q(x) \in \mathbb{Z}[x],\ k \in \mathbb{Z}\} = (x - a, 12).$$

Here $(x - a, 12)$ is the ideal generated by two elements. A general term $12r(x)$ can be put in the displayed form using $r(x) = r(a) + (x - a)s(x)$. The first isomorphism theorem gives

$$\mathbb{Z}[x]/(x - a, 12) \cong \mathbb{Z}_{12}.$$

With codomain $\mathbb{Z}$, no reduction is taken. The image is $\mathbb{Z}$, and the kernel is exactly $(x - a)$, so

$$\mathbb{Z}[x]/(x - a) \cong \mathbb{Z}.$$

These kernels differ. The constant 12 belongs to the first but not the second. Arbitrarily choosing a representative of a $\mathbb{Z}_{12}$ class does not define an inverse homomorphism to $\mathbb{Z}$, since $[0] = [12]$ would give different integer outputs.

3

The Euclidean algorithm gives

$$43 = 2 \cdot 17 + 9, \quad 17 = 1 \cdot 9 + 8, \quad 9 = 1 \cdot 8 + 1.$$

Back substitution gives

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Thus $17^{-1} \equiv -5 \equiv 38 \pmod{43}$, and

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$$x \equiv 5 \cdot 38 = 190 \equiv 18 \pmod{43}.$$

4

The first congruence gives $x = 1 + 4t$. Modulo 5,

$$1 + 4t \equiv 2, \qquad 4t \equiv 1, \qquad t \equiv 4 \pmod{5}.$$

Write $t = 4 + 5s$, so $x = 17 + 20s$. Modulo 7,

$$17 + 20s \equiv 3, \qquad 3 + 6s \equiv 3, \qquad s \equiv 0 \pmod{7}.$$

Therefore $x \equiv 17 \pmod{140}$, and the least nonnegative solution is 17.

5

The set $I = 2\mathbb{Z}$ is an additive subgroup and absorbs integer multiplication, since $r(2k) = 2(rk) \in I$. It is an ideal of $\mathbb{Z}$.

Likewise, $J = \{[0], [2], [4]\}$ is an additive subgroup of $\mathbb{Z}_6$ absorbing multiplication, hence an ideal. It has 3 elements, so the quotient has $6/3 = 2$ elements. Its cosets are J and $1 + J = \{[1], [3], [5]\}$. Their addition and multiplication agree with residues modulo 2, giving

$$\mathbb{Z}_6/J \cong \mathbb{Z}_2.$$

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6

The contrapositive of the defining zero-product property of an integral domain is $a, b \neq 0 \implies ab \neq 0$. A counterexample is $\mathbb{Z}_6$, where $[2], [3] \neq [0]$ but $[2][3] = [0]$.

7

Since $\gcd(10, 26) = 2$ divides 14, divide by 2 to obtain $5x \equiv 7 \pmod{13}$. As $5 \cdot 8 = 40 \equiv 1$,

$$x \equiv 7 \cdot 8 = 56 \equiv 4 \pmod{13}.$$

The two solutions modulo 26 are $4, 17$. Substitution gives $40 \equiv 14$ and $170 \equiv 14 \pmod{26}$.

For the system, $\gcd(10, 15) = 5$ divides $8 - 3 = 5$, so it is compatible. Writing $x = 3 + 10k$ gives $10k \equiv 5 \pmod{15}$, hence $2k \equiv 1 \pmod{3}$, $k \equiv 2 \pmod{3}$. Thus

$$x \equiv 23 \pmod{30}.$$

The uniqueness modulus is $\operatorname{lcm}(10, 15) = 30$. Modulo 150, there are five different solution classes, $23, 53, 83, 113, 143$, so uniqueness modulo 150 would be false.

Rings and modular arithmetic solutions (continued)

If the second residue is 9, the difference is 6, not divisible by 5, so no solution exists. The first congruence forces residue 3 modulo 5, while the second forces residue 4, a direct contradiction.

8

Additivity forces $\varphi([a]_4) = ae$, where $e = \varphi([1]_4)$, and $4e = 0 \pmod 6$. Multiplicativity forces $e^2 = \varphi(1^2) = e$. Conversely, these two conditions suffice. A change of representative by $4k$ changes the output by $4ke = 0$, proving well-definedness. Addition is preserved, and

$$\varphi(a)\varphi(b) = abe^2 = abe = \varphi(ab).$$

The condition $4e = 0 \pmod 6$ is equivalent to $3 \mid e$, so only $e = 0, 3$ are possible. Both are idempotent, since $0^2 = 0$ and $3^2 = 9 \equiv 3$. Thus all homomorphisms are the zero map and $\varphi([a]) = [3a]$.

The zero map has kernel $\mathbb{Z}_4$ and image $\{0\}$. The nonzero map has kernel $\{0, 2\}$ and image $\{0, 3\}$. Neither sends 1 to 1 of $\mathbb{Z}_6$, so there are no unital homomorphisms. The fact that natural reduction would require $6 \mid 4$ is consistent with this conclusion, but testing only that candidate would not classify all general homomorphisms.

9

The set S is nonempty and closed under subtraction and multiplication:

$$\operatorname{diag}(a, 0) - \operatorname{diag}(b, 0) = \operatorname{diag}(a - b, 0), \qquad \operatorname{diag}(a, 0)\operatorname{diag}(b, 0) = \operatorname{diag}(ab, 0).$$

It is a subring. Its own identity is $E = \operatorname{diag}(1, 0)$, which acts as identity on every member of S. The ambient identity $\operatorname{diag}(1, 1)$ is not in S, so it is not a unital subring of the ambient ring.

Take $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $E \in S$, but $AE = A \notin S$. Thus S is not an ideal. Closure only under products within S does not provide ambient absorption.

Since $\det P = 5 - 4 = 1$,

$$P^{-1} = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

Its entries are integers, and multiplication with P gives the identity, so P is a unit of $M_2(\mathbb{Z})$. Since $\det Q = 6 - 4 = 2$,

$$Q^{-1} = \frac{1}{2}\begin{pmatrix} 6 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -1 & 1/2 \end{pmatrix}.$$

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This is an inverse over $\mathbb{Q}$, but not an integer matrix. An integer inverse B would require $\det Q \det B = 1$, giving twice an integer equal to 1, impossible. In general an integer matrix is a unit of its integer matrix ring exactly when its determinant is ± 1. Necessity follows from determinants, and sufficiency follows from the adjugate formula.

10

Take $h(a) = a - 5$, with inverse $h^{-1}(y) = y + 5$. Then

$$h(a \oplus b) = a + b - 10 = (a - 5) + (b - 5) = h(a) + h(b),$$
$$h(a \otimes b) = ab - 5a - 5b + 25 = (a - 5)(b - 5) = h(a)h(b).$$

Thus h is an isomorphism to ordinary $\mathbb{Q}$, so the new structure is a field. Its zero is 5, its multiplicative identity is 6, and the additive inverse of a is $10 - a$. Exactly the elements $a \neq 5$ are invertible, with inverse

$$5 + \frac{1}{a - 5}.$$

For $a = 7$, this is $11/2$. Directly,

$$7 \otimes (11/2) = 77/2 - 35 - 55/2 + 30 = 6,$$

the new multiplicative identity. The ordinary numeral 0 is not the new additive zero; it is invertible since $h(0) = -5 \neq 0$.

11

The map $\Theta([x]_{18}) = ([x]_2, [x]_9)$ is well defined and preserves both operations. The residues $9, 10$ map to $(1, 0), (0, 1)$, respectively, so

$$\Theta^{-1}(a, b) = [9a + 10b]_{18}.$$

Reduction modulo each modulus verifies that this really is the inverse, proving the isomorphism.

An idempotent in the product satisfies $a^2 = a, b^2 = b$ coordinatewise. In $\mathbb{Z}_2$, $a = 0, 1$. In $\mathbb{Z}_9$, $9 \mid b(b - 1)$. Consecutive integers are coprime, so the factor 3^2 cannot split between them; it divides one whole factor. Thus $b \equiv 0$ or $1 \pmod 9$. The four pairs $(0, 0), (1, 0), (0, 1), (1, 1)$ give exactly

$$x = 0, 9, 10, 1.$$

This proves completeness without enumerating all residues.

Only 1 among them is a unit: cancellation of a unit in $x^2 = x$ forces $x = 1$. The nonzero elements $9, 10$ are zero divisors because $9 \cdot 10 = 90 \equiv 0 \pmod{18}$. By this chapter's convention, 0 is not listed as a nonzero zero divisor.

Rings and modular arithmetic solutions (continued)

For a unit in the product, the $\mathbb{Z}_2$ coordinate must be 1, and the $\mathbb{Z}_9$ coordinate can be $1, 2, 4, 5, 7, 8$. There are $1 \cdot 6 = 6$ units. Reconstructing gives $1, 5, 7, 11, 13, 17$, precisely the residues satisfying $\gcd(x, 18) = 1$.

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Rings and modular arithmetic solutions (continued)

Verification code appendix: modular_arithmetic.py

```
"""Exact integer algorithms for the chapter's congruence and CRT examples."""
from math import gcd

def extended_gcd(a: int,b: int):
    old_r,r=a,b;old_s,s=1,0;old_t,t=0,1
    while r:
        q=old_r//r
        old_r,r=r,old_r-q*r
        old_s,s=s,old_s-q*s
        old_t,t=t,old_t-q*t
    sign=1 if old_r>=0 else -1
    return sign*old_r,sign*old_s,sign*old_t

def linear_congruence(a: int,b: int,n: int):
    if n<=0:raise ValueError("modulus must be positive")
    d=gcd(a,n)
    if b%d:return []
    period=n//d
    if period==1:base=0
    else:
        _,inverse,_=extended_gcd(a//d,period)
        base=(inverse*(b//d))%period
    return [base+k*period for k in range(d)]

def combine_congruences(a: int,m: int,b: int,n: int):
    if m<=0 or n<=0:raise ValueError("moduli must be positive")
    g,inverse,_=extended_gcd(m,n)
    if (b-a)%g:return None
    period=m*(n//g)
    k=(inverse*((b-a)//g))%(n//g)
    return (a+m*k)%period,period

def power_mod(a: int,exponent: int,modulus: int):
    if exponent<0 or modulus<=0:raise ValueError("exponent must be nonnegative and modulus
positive")
    result=1%modulus
    for bit in bin(exponent)[2:]:
        result=(result*result)%modulus
        if bit=='1':result=(result*a)%modulus
    return result

if __name__=='__main__':
    assert linear_congruence(6,8,14)==[6,13]
    assert linear_congruence(6,9,14)==[]
    assert combine_congruences(5,8,9,12)==(21,24)
    assert combine_congruences(5,8,10,12) is None
    assert power_mod(5,13,17)==3
    print('modular arithmetic examples: PASS')
```

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Rings and modular arithmetic solutions (continued)

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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